



Faculty of Engineering and Technology
Department of Electrical and Computer Engineering
ENCS3320-Computer Networks

Task -UDP Client-Server Trivia Game Using Socket Programming

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Abstract

Imagine playing a trivia game where multiple players connect to a server, answer questions, and compete for the top score all in real time. This is made possible using UDP, a fast and lightweight protocol designed for quick communication. Unlike other protocols like TCP, which ensures every packet of data arrives safely, UDP focuses on speed, making it perfect for games where every second counts. The server acts as the game host, sending questions to all players and keeping track of their answers and scores. Players, as clients, receive the questions, submit their responses, and wait for updates from the server. This task shows how UDP and socket programming come together to create a fun and interactive multiplayer experience, while balancing speed and reliability in a distributed environment.

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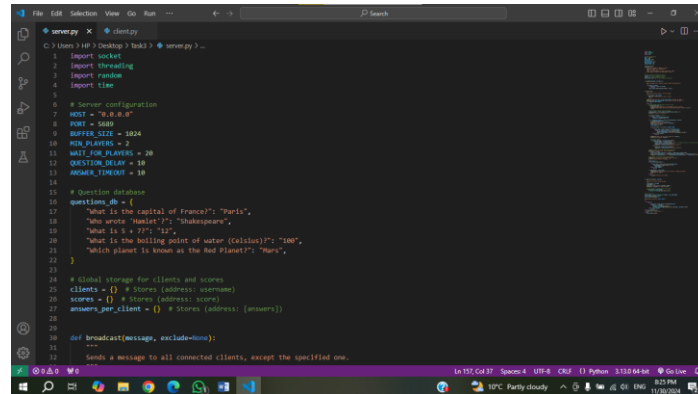
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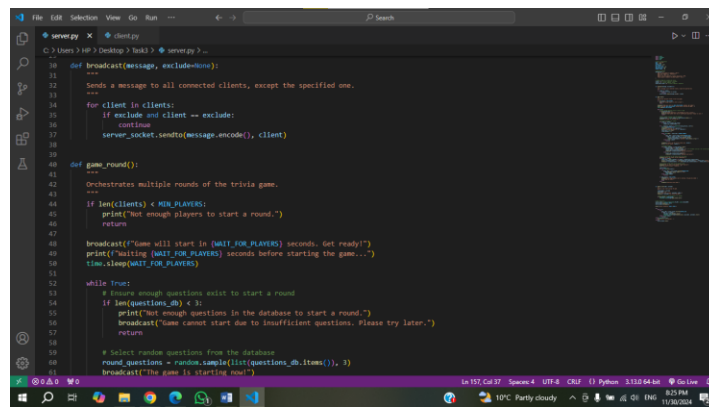
Codes & Discussion:

- The server code:



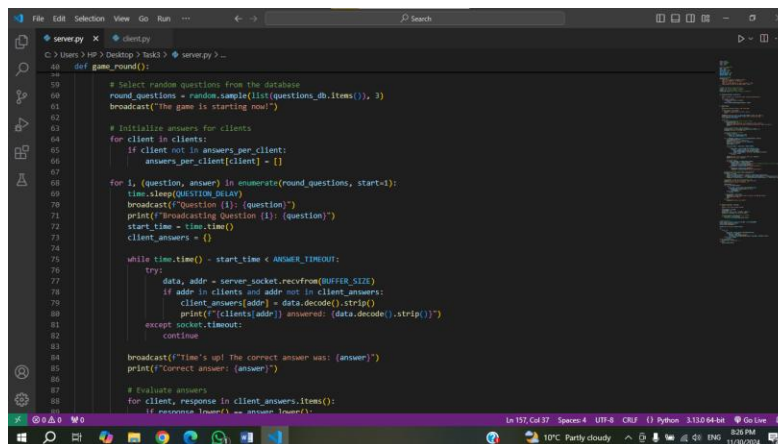
```
1 import socket
2 import threading
3 import random
4 import time
5
6 # Server configuration
7 HOST = "0.0.0.0"
8 PORT = 8080
9 BUFFER_SIZE = 1024
10 MIN_PLAYERS = 3
11 WAIT_FOR_PLAYERS = 20
12 QUESTION_DELAY = 10
13 ANSWER_TIMEOUT = 10
14
15 # Question database
16 questions_db = {
17     "What is the capital of France?": "Paris",
18     "Who wrote 'Hamlet'?": "Shakespeare",
19     "What is 5 + 7?": "12",
20     "What is the boiling point of water (Celsius)?": "100",
21     "Which planet is known as the Red Planet?": "Mars",
22 }
23
24 # Global storage for clients and scores
25 clients = {} # Stores (address: username)
26 scores = {} # Stores (address: score)
27 answers_per_client = {} # Stores (address: [answers])
28
29 def broadcast(message, exclude=None):
30     """
31     Sends a message to all connected clients, except the specified one.
32     """
```

Figure 1: server code



```
33
34     for client in clients:
35         if exclude and client == exclude:
36             continue
37         server_socket.sendto(message.encode(), client)
38
39 def game_round():
40     """
41     Orchestrates multiple rounds of the trivia game.
42     """
43     if len(clients) < MIN_PLAYERS:
44         print("Not enough players to start a round.")
45         return
46
47     broadcast(f"Game will start in {WAIT_FOR_PLAYERS} seconds. Get ready!")
48     print(f"Waiting {WAIT_FOR_PLAYERS} seconds before starting the game...")
49     time.sleep(WAIT_FOR_PLAYERS)
50
51     while True:
52         # Ensure enough questions exist to start a round
53         if len(questions_db) < 3:
54             print("Not enough questions in the database to start a round.")
55             broadcast("Game cannot start due to insufficient questions. Please try later.")
56             return
57
58         # Select random questions from the database
59         round_questions = random.sample(list(questions_db.items()), 3)
60         broadcast(f"The game is starting now!")
```

Figure 2: server code



```
61
62     # Initialize answers per client
63     for client in clients:
64         if client not in answers_per_client:
65             answers_per_client[client] = []
66
67     for i, (question, answer) in enumerate(round_questions, start=1):
68         time.sleep(QUESTION_DELAY)
69         broadcast(f"Question ({i}): {question}")
70         print(f"Broadcasting Question ({i}): {question}")
71         start_time = time.time()
72         client_answers = {}
73
74         while time.time() - start_time < ANSWER_TIMEOUT:
75             try:
76                 data, addr = server_socket.recvfrom(BUFFER_SIZE)
77                 if addr in clients and addr not in client_answers:
78                     client_answers[addr] = data.decode().strip()
79                     print(f"{clients[addr]} answered: {data.decode().strip()}")
80             except socket.timeout:
81                 continue
82
83         broadcast(f"Time's up! The correct answer was: {answer}")
84         print(f"Correct answer: {answer}")
85
86     # Evaluate answers
87     for client, response in client_answers.items():
88         if response.lower() == answer.lower():
```

Figure 3: server code

```

100 def game_round():
101     # Evaluate answers
102     for client, response in client_answers.items():
103         if response.lower() == answer.lower():
104             scores[client] = scores.get(client, 0) + 1 # Increment score by 1 for correct answer
105             answers_per_client[client].append("1")
106             broadcast(f"{client(client)} answered correctly")
107         else:
108             answers_per_client[client].append("0")
109             broadcast(f"{client(client)} answered incorrectly")
110
111     # After all questions in the round, print the scores
112     broadcast(f"End of the round! Current leaderboard:")
113     for client in clients:
114         total_score = sum(int(answer) for answer in answers_per_client[client])
115         print(f"{client(client)} ({' ' * 10} {total_score} points)")
116         broadcast(f"{client(client)} ({' ' * 10} {total_score} points)")
117
118     # Ask players if they want to play another round
119     broadcast("Do you want to play another round (yes/no)")
120     print("Waiting for players' responses...")
121
122     ready_players = 0
123     for _ in range(len(clients)):
124         try:
125             data, addr = server_socket.recvfrom(BUFFER_SIZE)
126             if data.decode().strip().lower() == "yes":
127                 ready_players += 1
128             except socket.timeout:
129                 continue
130
131

```

Figure 4: server code

```

132 def game_round():
133     if ready_players < MIN_PLAYERS:
134         broadcast("Not enough players want to continue. The game has ended.")
135         print("Game has ended.")
136         break
137     else:
138         broadcast("Starting a new round!")
139
140
141 def handle_client(addr, username):
142     """
143     handles a new client joining the game.
144     """
145     clients[addr] = username
146     scores[addr] = 0
147     print(f"New client joined: {username} ({addr})")
148     broadcast(f"{username} has joined the game!")
149
150     # Start the game if minimum players are reached
151     if len(clients) >= MIN_PLAYERS:
152         threading.Thread(target=game_round).start()
153
154
155 # Start the server
156 server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
157 server_socket.bind((HOST, PORT))
158 server_socket.settimeout(1)
159 print(f"Server started at {HOST}:{PORT}")
160

```

Figure 5: server code

```

161 def handle_client(addr, username):
162     scores[addr] = 0
163     print(f"New client joined: {username} ({addr})")
164     broadcast(f"{username} has joined the game!")
165
166     # Start the game if minimum players are reached
167     if len(clients) >= MIN_PLAYERS:
168         threading.Thread(target=game_round).start()
169
170
171 # Start the server
172 server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
173 server_socket.bind((HOST, PORT))
174 server_socket.settimeout(1)
175 print(f"Server started at {HOST}:{PORT}")
176
177 try:
178     while True:
179         try:
180             data, addr = server_socket.recvfrom(BUFFER_SIZE)
181             if addr not in clients:
182                 username = data.decode().strip()
183                 threading.Thread(target=handle_client, args=(addr, username)).start()
184             except socket.timeout:
185                 continue
186         except KeyboardInterrupt:
187             print("Server shutting down...")
188             finally:
189                 server_socket.close()
190

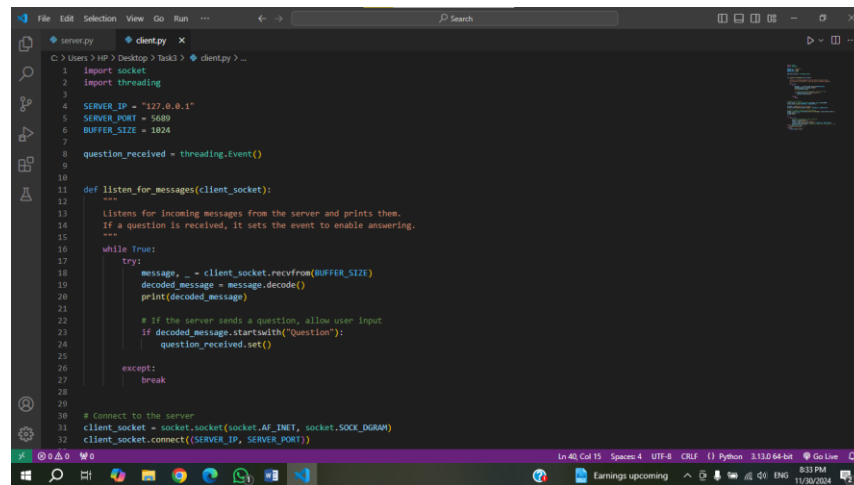
```

Figure 6: server code

- **Server code discussion:**

This Python code creates a multiplayer trivia game server using UDP. The server listens for players to join the game by connecting on port 5689, where each player is asked to provide a username. Once at least two players are connected, the server kicks off the game by randomly selecting three trivia questions from a pre-made list and sending them to all players. Each player has a set amount of time to answer, and the server collects their responses, checks if they're correct, and awards points. After each round, the server shows the current scores and asks if the players want to continue with another round. If not, enough players want to keep going, the game ends. The server uses threading to manage multiple players at once, ensuring the game runs smoothly for everyone.

- **The client code:**

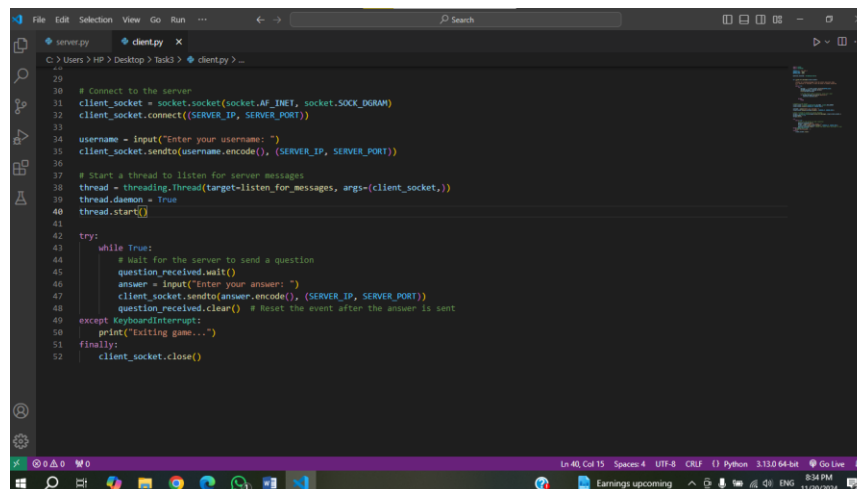


```

1  import socket
2  import threading
3
4  SERVER_IP = "127.0.0.1"
5  SERVER_PORT = 5480
6  BUFFER_SIZE = 1024
7
8  question_received = threading.Event()
9
10
11 def listen_for_messages(client_socket):
12     """
13     Listens for incoming messages from the server and prints them.
14     If a question is received, it sets the event to enable answering.
15     """
16     while True:
17         try:
18             message = client_socket.recvfrom(BUFFER_SIZE)
19             decoded_message = message.decode()
20             print(decoded_message)
21
22             # If the server sends a question, allow user input
23             if decoded_message.startswith("Question"):
24                 question_received.set()
25
26         except:
27             break
28
29 # Connect to the server
30 client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
31 client_socket.connect((SERVER_IP, SERVER_PORT))

```

Figure 7: the client code



```

29
30 # Connect to the server
31 client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
32 client_socket.connect((SERVER_IP, SERVER_PORT))
33
34 username = input("Enter your username: ")
35 client_socket.sendto(username.encode(), (SERVER_IP, SERVER_PORT))
36
37 # Start a thread to listen for server messages
38 thread = threading.Thread(target=listen_for_messages, args=(client_socket,))
39 thread.daemon = True
40 thread.start()
41
42 try:
43     while True:
44         # Wait for the server to send a question
45         question_received.wait()
46         answer = input("Enter your answer: ")
47         client_socket.sendto(answer.encode(), (SERVER_IP, SERVER_PORT))
48         question_received.clear() # Reset the event after the answer is sent
49 except KeyboardInterrupt:
50     print("Exiting game...")
51 finally:
52     client_socket.close()

```

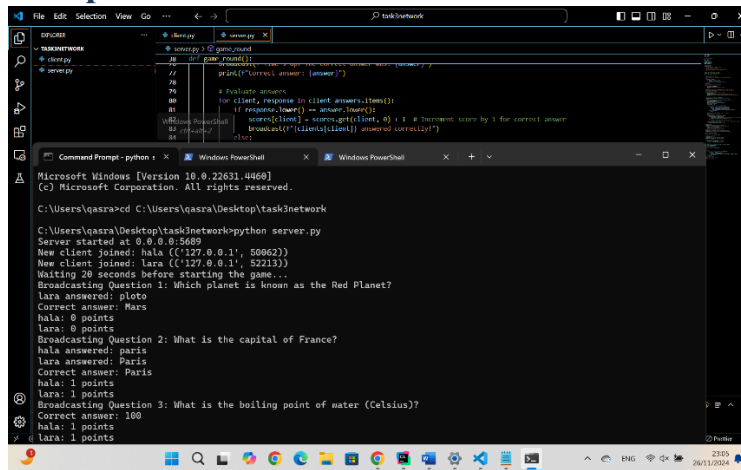
Figure 8: the client code

- **Client code discussion:**

This Python code is for the client side of a trivia game, where the client connects to the server using UDP. After the client connects, it asks the user to enter a username, which is then sent to the server. A separate thread listens for messages from the server and prints them out. When the server sends a trivia question, it lets the user know that they can now answer by triggering an event. The client then waits for the question, prompts the user to type their answer, and sends the response back to the server. After each answer, the event is reset so the user can only answer one question at a time. The game continues until the user decides to exit.

Results and Discussion:

- The first example:

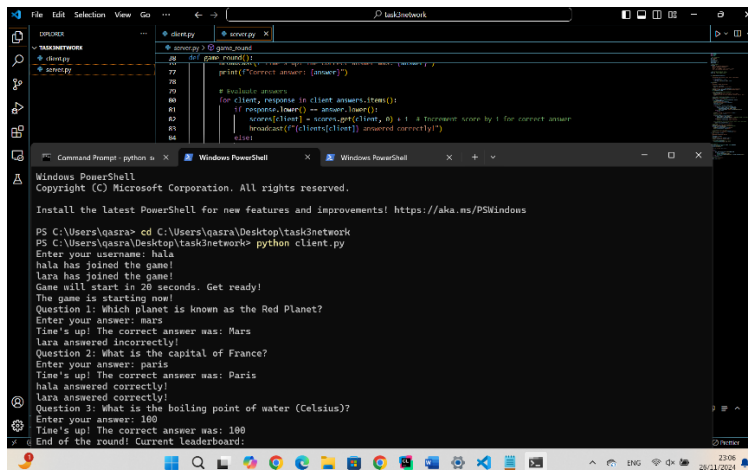


```
File Edit Selection View Go ... tasknetwork
tasknetwork
server.py
def game_round():
    # Broadcast question to server
    broadcast("Broadcasting Question 1: Which planet is known as the Red Planet?")
    # Evaluate answers
    for client, response in client_answers.items():
        if response.lower() == answer.lower():
            scores[client] = scores.get(client, 0) + 1 # Increase score by 1 for correct answer
            broadcast(f"{client[client]} answered correctly!")
        else:
            broadcast(f"{client[client]} answered incorrectly!")

Microsoft Windows [Version 10.0.22631.4609]
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C:\Users\qasra>cd C:\Users\qasra\Desktop\tasknetwork
C:\Users\qasra\Desktop\tasknetwork>python server.py
Server started at: 0.0.0.0:5689
New client joined: hala (('127.0.0.1', 50862))
New client joined: lara (('127.0.0.1', 52213))
Waiting 20 seconds before starting the game...
Broadcasting Question 1: Which planet is known as the Red Planet?
lara answered: plato
Correct answer: Mars
hala: 0 points
lara: 0 points
Broadcasting Question 2: What is the capital of France?
hala answered: paris
lara answered: Paris
Correct answer: Paris
hala: 1 points
lara: 1 points
Broadcasting Question 3: What is the boiling point of water (Celsius)?
Correct answer: 100
hala: 1 points
lara: 1 points
```

Figure 9: the output of server Ex1



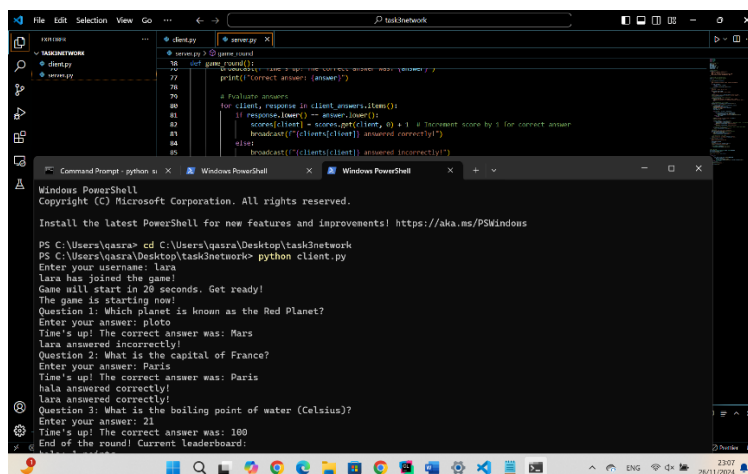
```
File Edit Selection View Go ... tasknetwork
tasknetwork
client.py
def game_round():
    # Broadcast question to server
    broadcast("Broadcasting Question 1: Which planet is known as the Red Planet?")
    # Evaluate answers
    for client, response in client_answers.items():
        if response.lower() == answer.lower():
            scores[client] = scores.get(client, 0) + 1 # Increase score by 1 for correct answer
            broadcast(f"{client[client]} answered correctly!")
        else:
            broadcast(f"{client[client]} answered incorrectly!")

Windows PowerShell
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PS C:\Users\qasra> cd C:\Users\qasra\Desktop\tasknetwork
PS C:\Users\qasra\Desktop\tasknetwork> python client.py
Enter your username: hala
hala has joined the game!
lara has joined the game!
Game will start in 20 seconds. Get ready!
The game is starting now!
Question 1: Which planet is known as the Red Planet?
Enter your answer: mars
Time's up! The correct answer was: Mars
lara answered incorrectly!
Question 2: What is the capital of France?
Enter your answer: paris
Time's up! The correct answer was: Paris
hala answered correctly!
lara answered correctly!
Question 3: What is the boiling point of water (Celsius)?
Enter your answer: 100
Time's up! The correct answer was: 100
End of the round! Current leaderboard:
```

Figure 10: the output of client1 Ex1



```
File Edit Selection View Go ... tasknetwork
tasknetwork
client.py
def game_round():
    # Broadcast question to server
    broadcast("Broadcasting Question 1: Which planet is known as the Red Planet?")
    # Evaluate answers
    for client, response in client_answers.items():
        if response.lower() == answer.lower():
            scores[client] = scores.get(client, 0) + 1 # Increase score by 1 for correct answer
            broadcast(f"{client[client]} answered correctly!")
        else:
            broadcast(f"{client[client]} answered incorrectly!")

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PS C:\Users\qasra> cd C:\Users\qasra\Desktop\tasknetwork
PS C:\Users\qasra\Desktop\tasknetwork> python client.py
Enter your username: lara
lara has joined the game!
Game will start in 20 seconds. Get ready!
The game is starting now!
Question 1: Which planet is known as the Red Planet?
Enter your answer: plato
Time's up! The correct answer was: Mars
lara answered incorrectly!
Question 2: What is the capital of France?
Enter your answer: Paris
Time's up! The correct answer was: Paris
hala answered correctly!
lara answered correctly!
Question 3: What is the boiling point of water (Celsius)?
Enter your answer: 21
Time's up! The correct answer was: 100
End of the round! Current leaderboard:
```

Figure 11: the output of client2 EX1

- The second example:

```

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PS C:\Users\qasra> cd C:\Users\qasra\Desktop\task3network
PS C:\Users\qasra\Desktop\task3network> python server.py
Server started at 0.0.0.0:5689
New client joined: hala (('127.0.0.1', 64301))
New client joined: lara (('127.0.0.1', 63118))
Waiting 20 seconds before starting the game...
Broadcasting Question 1: What is 5 + 7?
hala answered: 12
lara answered: 17
Correct answer: 12
Broadcasting Question 2: What is the capital of France?
hala answered: Paris
lara answered: los angeles
Correct answer: Paris
Broadcasting Question 3: What is the boiling point of water (Celsius)?
Correct answer: 100
hala = 2 points
lara = 0 points

```

Figure 12: output of server EX2

```

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PS C:\Users\qasra> cd C:\Users\qasra\Desktop\task3network
PS C:\Users\qasra\Desktop\task3network> python client.py
Enter your username: hala
hala has joined the game!
lara has joined the game!
Game will start in 20 seconds. Get ready!
The game is starting now!
Question 1: What is 5 + 7?
Enter your answer: 12
Time's up! The correct answer was: 12
hala answered correctly!
lara answered incorrectly!
Question 2: What is the capital of France?
Enter your answer: Paris
Time's up! The correct answer was: Paris
hala answered correctly!
lara answered incorrectly!
Question 3: What is the boiling point of water (Celsius)?
Enter your answer: 9
Time's up! The correct answer was: 100
End of the round! Current leaderboard:
hala = 2 points
lara = 0 points

```

Figure 13: output of client1 EX2

```

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PS C:\Users\qasra> cd C:\Users\qasra\Desktop\task3network
PS C:\Users\qasra\Desktop\task3network> python client.py
Enter your username: lara
lara has joined the game!
Game will start in 20 seconds. Get ready!
The game is starting now!
Question 1: What is 5 + 7?
Enter your answer: 17
Time's up! The correct answer was: 12
hala answered correctly!
lara answered incorrectly!
Question 2: What is the capital of France?
Enter your answer: los angeles
Time's up! The correct answer was: Paris
hala answered correctly!
lara answered incorrectly!
Question 3: What is the boiling point of water (Celsius)?
Enter your answer: 6
Time's up! The correct answer was: 100
End of the round! Current leaderboard:
hala = 2 points
lara = 0 points

```

Figure 14: output of client2 EX2

- More than one round example:

```

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PS C:\Users\qasra> cd C:\Users\qasra\Desktop\task3network
PS C:\Users\qasra\Desktop\task3network> python server.py
Server started at 0.0.0.0:5689
New client joined: hala ([127.0.0.1], 56912)
New client joined: lara ([127.0.0.1], 57540)
Waiting 20 seconds before starting the game...
Broadcasting Question 1: Who wrote 'Hamlet'?
Correct answer: Shakespeare
Broadcasting Question 2: Which planet is known as the Red Planet?
lara answered: ploto
Correct answer: Mars
Broadcasting Question 3: What is the capital of France?
hala answered: paris
lara answered: lo
Correct answer: Paris
hala (1) = 1 points
lara (0+0) = 0 points
Waiting for players' responses...
Broadcasting Question 1: Who wrote 'Hamlet'?
lara answered: shakespeare
Correct answer: Shakespeare
Broadcasting Question 2: What is the boiling point of water (Celsius)?
lara answered: 100
Correct answer: 100
Broadcasting Question 3: What is 5 + 7?
hala answered: 12
lara answered: 7
Correct answer: 12
hala (1+1) = 2 points
lara (0+0+0+1+0) = 1 points
Waiting for players' responses...
Broadcasting Question 1: Who wrote 'Hamlet'?
Correct answer: Shakespeare
Broadcasting Question 2: Which planet is known as the Red Planet?

```

Figure 15: output of server EX3

```

PS C:\Users\qasra\Desktop\task3network> python client.py
Enter your username: hala
hala has joined the game!
lara has joined the game!
The game is starting now!
Game will start in 20 seconds. Get ready!
Question 1: Who wrote 'Hamlet'?
Enter your answer: shakTime's up! The correct answer was: Shakespeare
Question 2: Which planet is known as the Red Planet?
lara answered incorrectly!
Time's up! The correct answer was: Mars
lara answered incorrectly!
Question 3: What is the capital of France?
Enter your answer: paris
Time's up! The correct answer was: Paris
lara answered correctly!
lara answered incorrectly!
End of the round! Current leaderboard:
hala (1) = 1 points
lara (0+0) = 0 points
Do you want to play another round? (yes/no)
Starting a new round!
The game is starting now!
Question 1: Who wrote 'Hamlet'?
Enter your answer: me
Time's up! The correct answer was: Shakespeare
lara answered incorrectly!
Question 2: What is the boiling point of water (Celsius)?
Enter your answer: 4
Time's up! The correct answer was: 100
lara answered correctly!
Question 3: What is 5 + 7?
Enter your answer: 12
Time's up! The correct answer was: 12
lara answered correctly!
lara answered incorrectly!
End of the round! Current leaderboard:

```

Figure 16: output of client1 EX3

```

Starting a new round!
The game is starting now!
Question 1: What is 5 + 7?
Time's up! The correct answer was: 12
Question 2: Who wrote 'Hamlet'?
Time's up! The correct answer was: Shakespeare
Question 3: What is the capital of France?
Time's up! The correct answer was: Paris
End of the round! Current leaderboard:
hala (1+1+1+1) = 4 points
lara (0+0+0+1+0) = 1 points
Do you want to play another round? (yes/no)
Starting a new round!
The game is starting now!
Question 1: Who wrote 'Hamlet'?
Time's up! The correct answer was: Shakespeare
Question 2: Which planet is known as the Red Planet?
Time's up! The correct answer was: Mars
Question 3: What is 5 + 7?
Time's up! The correct answer was: 12
End of the round! Current leaderboard:
hala (1+1+1+1) = 4 points
lara (0+0+0+1+0) = 1 points
Do you want to play another round? (yes/no)
Starting a new round!
The game is starting now!
Question 1: Which planet is known as the Red Planet?
Time's up! The correct answer was: Mars
Question 2: What is the capital of France?
Time's up! The correct answer was: Paris
Question 3: What is the boiling point of water (Celsius)?
Time's up! The correct answer was: 100
End of the round! Current leaderboard:
hala (1+1+1+1) = 4 points
lara (0+0+0+1+0) = 1 points
Do you want to play another round? (yes/no)

```

Figure 16: output of client2 EX3

- **Discussion:**

When the server starts, it will let everyone know that it's up and running, waiting for players on port 5689. As players join, it will show their username and IP address. Once enough players are in, the server will announce that the game is starting and send out the first question. It will print the question for everyone to see, and as players respond, the server will display each answer, along with whether it was right or wrong. At the end of each round, the server will share the scores, showing how each player did. If players want to continue, it will ask if they're ready for another round, but if not, enough players are interested, the game will end. When the server shuts down, it will let everyone know it's closing. On the client side, when a player connects, they'll see a message confirming they're connected to the server. As the game progresses, the client will display messages about the game starting, the questions being asked, and updates to the leaderboard. The client will prompt the player for an answer when a question arrives, and once the answer is submitted, it will confirm that the answer was sent. At the end of the round, the client will show the updated scores and any disconnection messages if the game is over or the server shuts down.

Conclusion

In conclusion, UDP makes real-time trivia games fast and exciting by prioritizing speed over perfect reliability. With the server hosting questions and tracking scores, and players submitting answers in real-time, the game stays interactive and engaging. This setup shows how clever use of technology can bring people together for a fun and competitive multiplayer experience.